

Unit 4, Lesson 4: Plotting, Ordering, and Comparing Rational Numbers – Part 1

Benchmark

MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.

Learning Target

- ☐ I can plot, compare, and order fractional representations of rational numbers.

4.4.1 Warm-Up: Closer to Zero

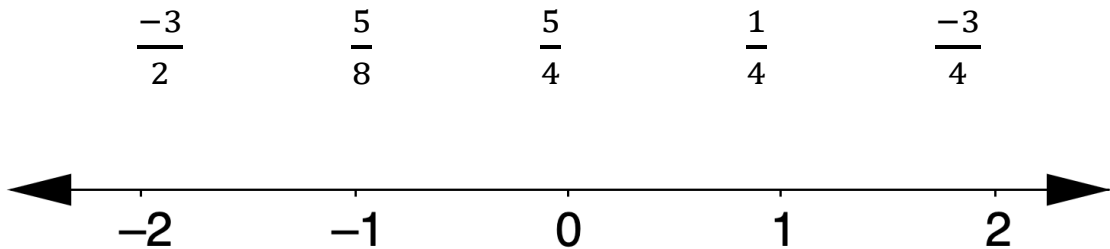
1. For each pair of numbers, determine which value is closest to 0.

Pair	Closest to Zero
$\frac{9}{11}$ or $\frac{15}{11}$	
$\frac{1}{5}$ or $\frac{1}{9}$	
1.25 or $\frac{5}{4}$	
0.01 or 0.001	

Discuss with your partner the strategy you used.

4.4.2 Exploration: Plotting and Comparing Fractions

1. Plot and label the rational numbers on the number line below.



Using the rational numbers plotted above, answer the following questions.

2. Andre says that $\frac{1}{4}$ is less than $-\frac{3}{4}$ because out of the two values, $\frac{1}{4}$ is closer to 0. Explain why you agree or disagree with Andre.

3. Complete the table for each pair of numbers.

Pair	Farthest Distance From 0	Greater Value
$\frac{1}{4}$ or $\frac{5}{4}$		
$-\frac{3}{4}$ or $\frac{5}{8}$		

4. Is the value that is farther from 0 always the greater value? Explain your reasoning.

4.4.3 Activity: Ordering Rational Number Cards

Your teacher will give you a set of number cards. Order them from least to greatest.

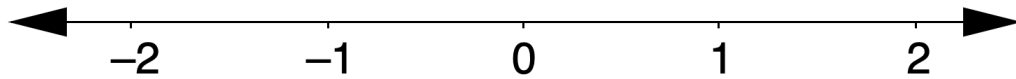
Your teacher will give you a second set of number cards. Add these values to the correct places in the first ordered set.

1. In the first round of ordering, how did ordering negative numbers compare to ordering positive numbers?
2. What strategies helped you determine how to order the rational numbers represented as a decimal or a percent after you had already ordered the fractions?

4.4.4 Your Turn!

1. Plot the following values on the number line. Label each point with its numeric value.

$$1.5 \quad -\frac{3}{2} \quad -\frac{4}{3} \quad 0.75 \quad \frac{3}{2}$$



2. Use the symbols $<$, $>$ or $=$ to compare each pair of values.

$$-\frac{4}{3} \quad \underline{\hspace{1cm}} \quad -\frac{3}{2}$$

$$0.75 \quad \underline{\hspace{1cm}} \quad \frac{3}{2}$$

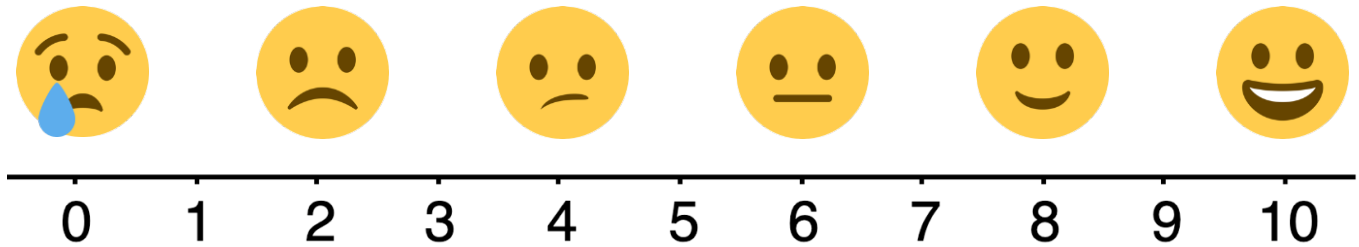
$$\frac{3}{2} \quad \underline{\hspace{1cm}} \quad 1.5$$

3. Order the set of rational numbers.

$$5.2 \quad -3\frac{1}{2} \quad 5\frac{1}{2} \quad -2\frac{3}{4}$$

4.4.5 Wrap-Up: Reflection

1. Circle the emoji that best represents your understanding of the lesson today.



2. Write an explanation of why you chose that emoji.